

Deep Learning for Atmospheric Composition Estimation and Earth-like Exoplanet Identification

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1 Introduction

With recent advancements in telescope technology, astronomers can observe farther and with greater resolution than ever before. Since 2009, starting with the Kepler Mission, NASA and private space companies have exploited these advancements to search the stars for exoplanets—planets orbiting stars outside our solar system. As of today, more than 5,800 exoplanets have been confirmed, out of the billions estimated to exist. With each confirmed exoplanet, we uncover another opportunity to find extraterrestrial life or a potential future home for humanity.

One of the crucial characteristics we can learn about an exoplanet is the composition of its atmosphere, assuming one exists, as it is a critical component in the planet’s ability to support life as we know it. Atmospheric composition is often studied through a technique called atmospheric spectroscopy. This process begins when starlight that has passed through an exoplanet’s atmosphere reaches a telescope. By analyzing the resulting spectrum, we can infer which compounds are present based on the absorption or emission features at specific wavelengths. Compounds produce characteristic emission or absorption lines, and the intensity of these features correlates with their relative abundance.

Exoplanets exhibiting spectral markers for oxygen, water vapor, and methane, for example, are of particular interest, as these compounds may signal the potential for hosting life.

Currently, determining the atmospheric composition of exoplanets relies on Bayesian Atmospheric Retrieval methods. This probabilistic approach estimates the most likely atmospheric parameters given observed spectra but requires prior probability distributions and complex forward models built using field expertise. These retrievals also demand significant computational resources and can take between hours and days to complete, depending on the complexity of the atmosphere and the signal-to-noise ratio of the observations.

In this work, we propose replacing Bayesian retrieval with a machine learning approach, developing a model that can directly predict the relative atmospheric abundances of key molecules from exoplanet spectra. Our model acts as a feature extractor for a downstream clustering algorithm, capable of prioritizing exoplanets for further study in the search for habitable worlds.

We build a convolutional neural network (CNN) capable of learning the complex relationships between spectral features and the relative abundances of compounds of interest, those being water, molecular oxygen, molecular nitrogen, carbon dioxide, and methane. Because there is currently an insufficient quantity of real, high-quality exoplanet spectra available to train large machine learning models, we train and test our model on a synthetic dataset composed of purely terrestrial (rocky) exoplanets, as gas giants are inhospitable to human life.

After predicting atmospheric abundances, we incorporate these predictions—along with planetary parameters such as radius and surface temperature (collectible via transit and radial velocity methods)—into an instance-based nearest-neighbors approach. This step allows us to efficiently prioritize exoplanets that merit further investigation with respect to the search for extraterrestrial life and human-compatible environments.

2 Related work

The dataset used to train and test our model consists of features and spectra for over three million synthetically generated terrestrial exoplanets. The method of its generation is detailed in the paper by Zorzan et al. [1], which introduces a machine learning-ready dataset for exoplanet atmospheric retrieval. Although the dataset is generated through retrieval-based simulations, our goal is to bypass such methods entirely during inference. Planetary parameters such as density, radius, and surface temperature were sampled uniformly within bounds defined by the “optimistically habitable” zone from Kopparapu et al. [2]. In addition to planetary characteristics, each synthetic exoplanet includes relative abundances for selected atmospheric compounds. These include water vapor, molecular nitrogen, molecular oxygen, methane, and carbon dioxide—molecules that are particularly relevant for assessing potential habitability. Notably, molecular nitrogen, oxygen, and water vapor account for nearly 99% of Earth’s atmosphere, making their modeled presence informative for life-supporting conditions. Other, less life-compatible compounds (e.g., sulfur dioxide or carbon monoxide) are also simulated to increase chemical diversity and capture a broader range of potential exoplanets.

The second component of the dataset consists of synthetic atmospheric spectra generated from the corresponding atmospheric compositions, one spectra per exoplanet. The presence of a molecule in an exoplanet’s atmosphere produces distinctive absorption or emission features within its spectrum. The dataset includes spectral signals for both the exoplanet and its host star, as well as the combined signal, for the range of 0–640 microns, with radiation intensity values that encode compositional information. We use both the isolated exoplanet

spectra and planetary features from a subset of the three million samples to train our CNN and downstream clustering model.

The conventional method for inferring atmospheric composition from spectra is Bayesian Atmospheric Retrieval. Lustig-Yaeger et al. [3] describe this process in the context of population-level studies of habitable zone exoplanets. This approach requires expert knowledge in planetary physics and relies on physical models such as the carbonate-silicate weathering feedback cycle, radiative transfer theory, climate equilibrium modeling, and prior probabilities for specific molecule abundances. While effective, Bayesian retrieval is computationally expensive and is less accessible to non-specialists.

In contrast, we propose a data-driven approach using supervised learning. By training a neural network to learn the mapping from spectra to molecular abundances, we can eliminate the need for expert-crafted generative models and reduce computational time substantially. Although our training data was generated using forward models (i.e., simulated retrievals), our trained model is designed to generalize and could, in principle, be applied directly to real observational spectra. Once trained, our method does not require any retrieval process at inference time, making it faster and potentially more scalable to large datasets from future telescopes.

2.1 Predicting atmospheric abundances from spectra

We identified five key molecules whose relative abundances in an atmosphere provide critical information about a planet’s potential habitability: H_2O , CO_2 , N_2 , O_2 , and CH_4 . These molecules were chosen due to their necessity for life and presence in Earth’s atmosphere. Predicting the relative abundance of these molecules is framed as a regression task, requiring the model to learn complex non-linear relationships between the radiation intensities and molecular presence in the atmosphere.

We used a randomly selected subset of the PSG dataset to train a deep neural network in a supervised setting to predict atmospheric characteristics from observed spectra. Specifically, the dataset was split into a training set (273,000 samples), a validation set (25,000 samples), and a test set (2,500 samples). Each spectrum was clipped at the 2-micron wavelength due to the simulated coronagraph extending only to the 2-micron wavelength. A coronagraph filters out the stellar signal and thus in its absence the planetary signal is heavily diluted. The spectrum could then be represented as a vector of length 4379, where each value corresponds to the radiation intensity at a specific wavelength with non-uniform deltas between adjacent entries.

Following Zorzan et al. (2025), we adopted their optimal model architecture: Conv1d(64)-tanh-MaxPool-Conv1d(64)-relu-MaxPool-Conv1d(128)-relu-MaxPool-Conv1d(256)-relu-FC(256)-relu-FC(12). The network utilizes the ADAM optimizer to minimize the mean squared error (MSE) loss defined as:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2,$$

where $\hat{y}_i$ and y_i are the predicted and true abundances for the i -th sample, and N is the total number of samples.

Training was conducted for up to 150 epochs, with early stopping implemented to prevent overfitting. The model progressively increases the number of filters in the convolutional layers (64, 128, 256) while applying max-pooling to reduce spatial dimensions, ultimately producing a 5-dimensional output vector corresponding to the predicted abundances.

To estimate uncertainty in the predictions, we employed Monte Carlo dropout during inference. For each test input, we performed 600 stochastic forward passes, each with random dropout applied with a rate of 0.1, and used the mean of the resulting output distribution as our final prediction.

2.2 Similarity-based pruning via nearest neighbors

After obtaining predicted molecular abundances, we sought to prioritize planets for further analysis by comparing their atmospheric and planetary properties to those of Earth. Rather than definitively classifying planets as "habitable" or "non-habitable," our aim was to identify a local neighborhood of planets whose conditions most closely resemble those conducive to Earth-like life, thus producing a targeted list of true prospects.

We employed a nearest-neighbors approach using Euclidean distance in feature space to assess similarity. Specifically, we used the `NearestNeighbors` method from `scikit-learn` to locate the k planets most similar to Earth based on their standardized feature vectors. This method does not produce explicit clusters; rather, it ranks all datapoints by proximity to a query point—in this case, Earth—within the chosen feature space.

The features used for comparison included predicted molecular abundances (H_2O , O_2 , CO_2 , CH_4 , N_2), as well as planetary surface temperature and radius. All features were standardized to zero mean and unit variance to ensure equal weighting across scales. We varied k to study how the similarity-based averages of nearby planets evolved with neighborhood size. Ultimately, we chose $k = 75$. This comes from the NASA estimates of 300 million habitable planets in our galaxy of the 100 billion total exoplanets. We then get a rate of 0.003, which when multiplied by our test set size of 25,000, yields 75. We don't expect to see many more habitable worlds than this, so this imposes a sort of upper bound.

Instead of producing discrete clusters, this method yields a continuous similarity spectrum. To characterize the "most Earth-like" region of the data, we computed the average atmospheric and planetary properties of the $k = 75$ closest neighbors to Earth. These averages serve as a statistical benchmark for habitability, representing planets that are most comparable to Earth in terms of atmospheric composition and basic planetary characteristics.

To contextualize the analysis, we inserted Earth as a synthetic datapoint into the dataset and measured its similarity to the rest of the exoplanet population. Planets that consistently appear in the top tier of similarity rankings may warrant further, more detailed atmospheric retrieval analysis, while those with little resemblance to Earth are deprioritized.

3 Results

3.1 Atmospheric abundance predictions

To evaluate our model’s ability to infer atmospheric composition from spectra, we focused on predicting the relative abundances of five molecules: H_2O , CO_2 , N_2 , O_2 , and CH_4 . These gases were selected for their relevance to assessing planetary habitability. The model demonstrated strong performance on four of the five, with an average MSE of 0.0683 and average R^2 of 0.9344, excluding N_2 .

Figure 1 shows predicted versus true abundances for each molecule across 2,500 test planets. Each data point represents the mean of 600 Monte Carlo dropout runs for a single exoplanet. The outputs cluster tightly around the diagonal, indicating reliable regression performance. An example of the distribution of predictions for H_2O and CH_4 for a singular test case is shown in Figure 2, where the predicted mean closely aligns with the ground truth despite the range of individual outputs. These plots also offer insight into the model’s lack of conviction at times for the true result value.

Compared to earlier models trained on 10,000 examples (our prior work) or the 110,000 used by Zorzan et al. [1], our model significantly improves performance. This improvement is largely due to the use of a 273,000-sample training set, which better supports the learning of our CNN with approximately 18 million trainable parameters. The model’s depth and parameter complexity require this scale to generalize effectively, given the subtle spectral cues involved in predicting molecular abundances.

3.2 Model limitations and additional molecule predictions

While our model performed well on most targets, predictions for N_2 were significantly less accurate. This trend is consistent with the findings in [1], where nitrogen estimates also clustered near the mean of the distribution rather than showing meaningful variation. This behavior is puzzling: N_2 values were drawn from the same distribution as O_2 and CO_2 , and exhibit comparable variance within the test set. One possibility is that spectral features associated with N_2 are weak or non-distinct in the wavelength range used (up to 2 microns), making learning difficult even with large datasets.

This poses a practical concern: nitrogen makes up roughly 80% of Earth’s atmosphere, so accurate N_2 estimation is critical for identifying Earth analogs. The model’s tendency to underpredict high- N_2 atmospheres and overpredict low- N_2 ones could distort downstream habitability assessments.

One potential workaround is to estimate N_2 by subtracting the predicted abundances of all other compounds from 1, as the model’s predictions are normalized. This constraint, while susceptible to propagation of errors in the other molecules, could yield a better prediction.

Despite the N_2 issue, the model generalized well to additional compounds simulated in the PSG dataset. Predictions for N_2O , NH_3 , NO_2 , O_3 , and SO_2 yielded an average MSE of 0.0618 and R^2 of 0.9374 — matching or slightly

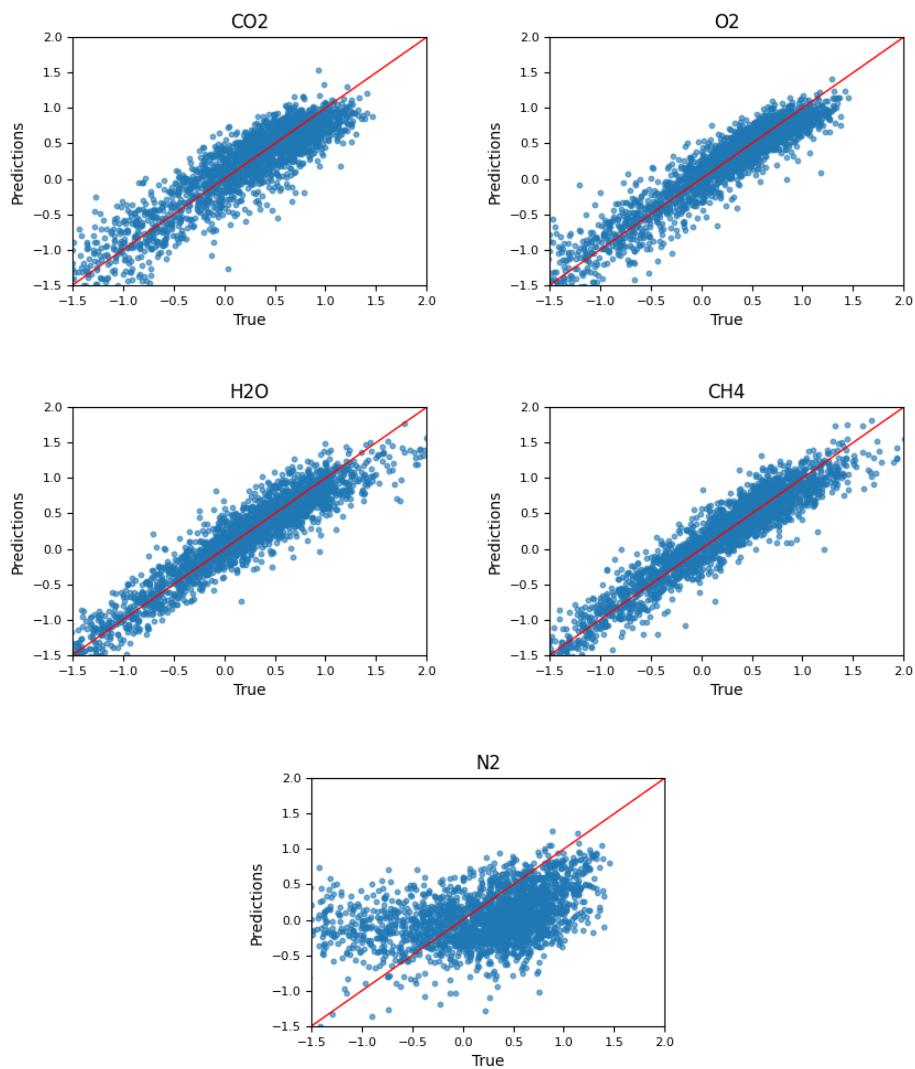


Figure 1: Scatterplots illustrating molecular abundance prediction performance for the logarithm of the normalized abundance of O₂, H₂O, CH₄, CO₂, and N₂ across 2,500 test planets. Each point is the average of 600 dropout runs. The diagonal line represents a perfect prediction.

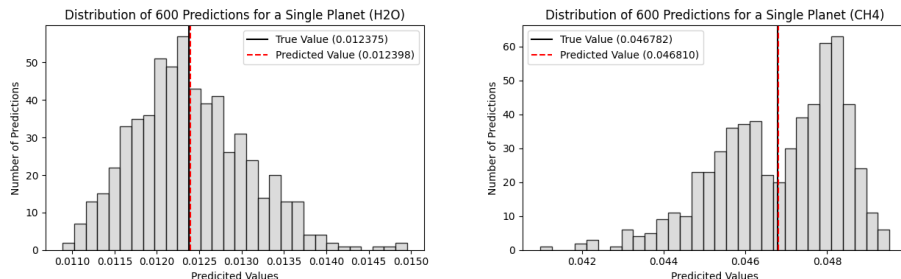


Figure 2: Distribution of 600 Monte Carlo dropout predictions for a single exoplanet’s H_2O and CH_4 abundances. The true value is shown as a solid line; the final model estimate is the mean (dotted line).

outperforming the results on our primary molecules. A second tier of molecules (C_2H_6 and CO) saw reduced performance, with MSE of 0.5361 and R^2 of 0.4650. These should be further investigated along with nitrogen.

3.3 Similarity-based filtering and prospective set reduction

Table 1: Average feature values of Earth’s 75 nearest neighbors

Feature	Mean Value	Earth Value
Planet Radius (km)	6289.51	6371
Surface Temp (K)	277.81	288
H_2O	0.0301	0.03
CO_2	0.0651	0.0004
O_2	0.1828	0.21
N_2	0.6748	0.78
CH_4	0.0141	0.00018

We used a similarity-based filtering strategy using a K-Nearest Neighbors (KNN) approach. By computing Euclidean distances in a standardized feature space, we identified the 75 planets most similar to Earth based on atmospheric composition and physical characteristics.

The averaged properties of these 75 nearest neighbors are shown in Table 1. This subset is tightly centered around Earth-like values, suggesting that nearest-neighbor techniques can effectively narrow a dataset to realistic habitability candidates. The mean planetary radius (6289.5 km) is just 81.5 km shy of Earth’s, and the average surface temperature (277.8 K) falls well within the plausible habitable range. These values reflect a preference for moderate planetary environments without bias toward extremes.

Atmospheric abundances further support the method’s utility. The average H_2O level (0.0301) nearly matches Earth’s (0.03), while oxygen (0.1828) and

nitrogen (0.6748) levels approach Earth’s atmospheric composition (0.21 and 0.78, respectively). The relatively elevated CO_2 and CH_4 values likely reflect the broader synthetic dataset distribution but still suggest moderately oxidizing atmospheres with trace greenhouse gases, consistent with early-Earth analogs. When more distant neighbors are considered, the values for these two features increases further and gets more distant from Earth. Similarly, nitrogen levels decrease away from an Earth-like average when more planets are taken into account.

Overall, this instance-based selection captures a concentrated pocket of Earth-like planets from a broader, largely uniform sample. By identifying planets most similar to a known habitable reference, we are able to reduce the candidate pool to 75 out of 25,000 planets (.003%)—a focused margin that respects both biological plausibility and resource constraints.

4 Conclusion

In this work, we developed and evaluated a CNN capable of learning the relationship between observed spectra and the abundances of atmospheric molecules in terrestrial exoplanets. Our model achieved strong predictive performance for four of the five primary atmospheric constituents, outperforming prior work on all molecules and demonstrating improved but poor performance on N_2 , C_2H_6 , and CO . This boosts the utility of deep learning models in the evaluation of exoplanet atmospheres.

To reduce the candidate search space for habitable worlds, we adopted a K-Nearest Neighbors approach using predicted molecular abundances and physical planet features. Rather than clustering all planets, this method identifies those most similar to Earth in feature space. The 75 closest neighbors revealed average values that closely mirrored Earth’s—matching its water and oxygen levels, approximating surface temperature and radius, and showing a preference for high nitrogen abundance. This suggests that instance-based learning can effectively isolate Earth analogs, even from a synthetically uniform dataset.

Future work should focus on improving predictions for poorly modeled molecules, particularly N_2 due to its importance to life on Earth, either through alternative model architectures or by using the requirement of normalization of the abundances. Expanding the training dataset further may also improve generalization given the model’s high parameter count. Once reliable prediction accuracy is achieved, a full pipeline integrating abundance estimation and nearest-neighbor filtering could be deployed to identify promising planets automatically. Additionally, the dataset should be expanded to include a broader range of exoplanet types to increase generalization.

We hope this work contributes to the growing effort to characterize exoplanet atmospheres and accelerates the search for potentially habitable worlds.

References

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Appendix A: Code Availability

The source code used in this study is publicly available at:

https://github.com/wilbrownstead/Comp562_Repo